

Towards Efficient Vision, Language and Multimodal Models

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The Era of Large Models

Introduction

- Deep learning drives unprecedented progress in various cognitive tasks
 - Modern models match or even surpass human performance¹

Question

Suppose that $X_1, X_2, \dots$ are real numbers between 0 and 1 that are chosen independently and uniformly at random. Let $S = \sum_{i=1}^k X_i/2^i$, where k is the least positive integer such that $X_k < X_{k-1}$, or $k = \infty$ if there is no such integer. Find the expected value of S .

Correct Answer: $2e^{1/2} - 3$

AI Models Answers

ChatGPT: $\mathbb{E}[S] = 2 - e^{1/2}$

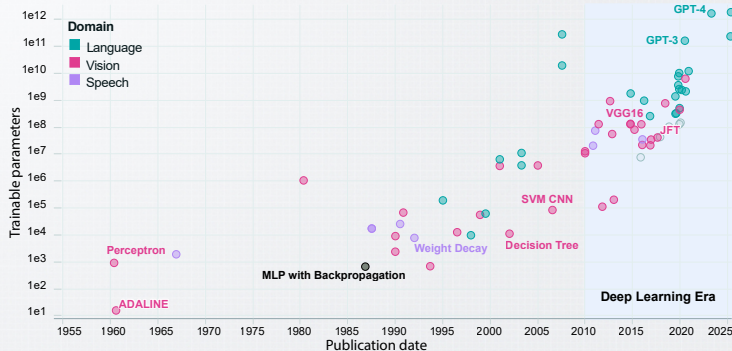
Gemini: The expected value of S is $\frac{1}{2} (e^{1/2} - 1)$

¹Bengio et al. *International Scientific Report on the Safety of Advanced AI*, 2025

Machine Learning Trends

Introduction

- Notable Machine Learning Models²
 - Influential models within the AI ecosystem
 - State-of-the-art advancements, historical significance, or high citation rates



¹<https://epoch.ai/data/ai-models>

²Morrison et al. *Holistically Evaluating the Environmental Impact of Creating Language Models*. ICLR, 2025

Advances in Deep Learning

Introduction

- Visual Models
 - Video generation
- Language Models
 - Natural Language Processing
- Large Vision-Language Models (LVLMs)
 - Image and text understanding

Story Summary: Hinton gives LeCun a task, which is not easy...

Input B/W Human Images



Generated dialogues post-edited by human

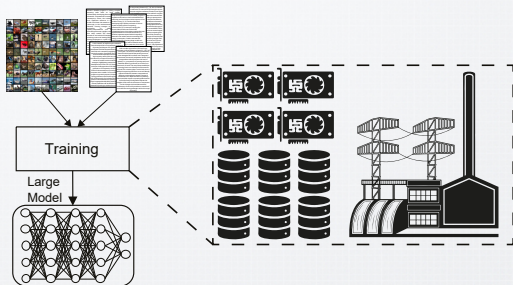


By DiffSensei¹

Energy Consumption

Introduction

- There is a growing demand for energy to power AI workloads¹
 - [...] cada sessão interativa com o modelo GPT-3 consome meio litro de água fresca²
 - O consumo de água para resfriamento dos data centers do Google [...] é comparável ao de uma cidade de 150 a 200 mil habitantes (Florianópolis com 500 mil habitantes, por exemplo, consome cerca de 25 bilhões de litros de água por ano)²



Model (#Params)	Carbon Emissions (tCO ₂ eq)	Equivalent to
< 1B	6	675 gal. of gasoline
1B	36	40 × NY ↔ SF
7B	65	150 oil barrels

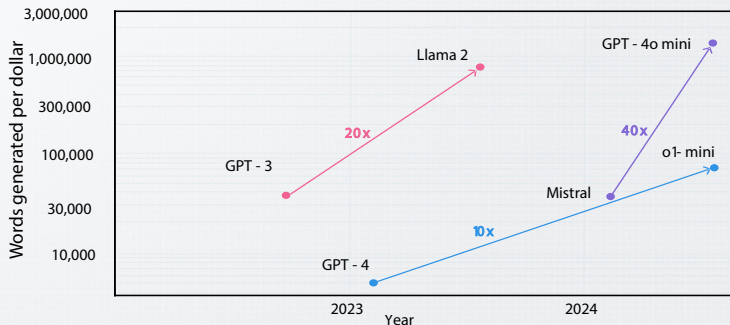
¹Morrison et al. *Holistically Evaluating the Environmental Impact of Creating Language Models*. ICLR, 2025

²IA para o Bem de Todos. Proposta de Plano Brasileiro de Inteligência Artificial 2024-2028

Financial Cost

Introduction

- Modern models are more efficient in terms of inference per cost¹
- The largest model providers are producing up to a hundred billion tokens per day²

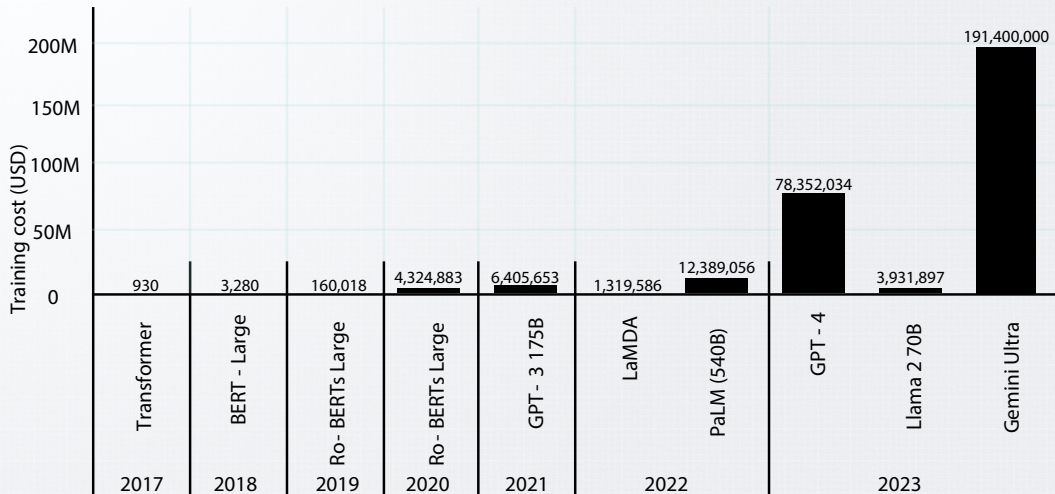


¹Bengio et al. *International Scientific Report on the Safety of Advanced AI*, 2025

²Morrison et al. *Holistically Evaluating the Environmental Impact of Creating Language Models*. ICLR, 2025

Financial Cost

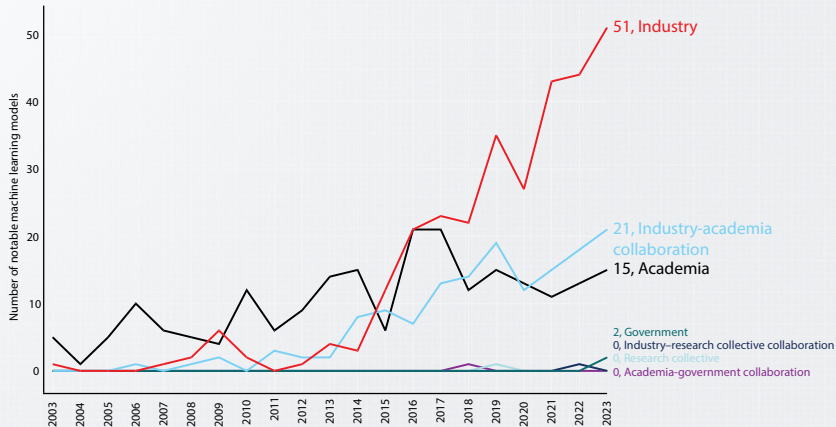
Introduction



Notable Models by Sector

Introduction

- Companies typically have more access to computational resources than academic institutions¹



¹Maslej et al. *Artificial Intelligence Index Report*, 2024

Research Projects

Introduction

- Scalable and Efficient Inference
 - Model Compression through Pruning
- Resource-Efficient Training
 - Low-rank Adaptation
 - Data selection and distillation
- Test-Time Compute

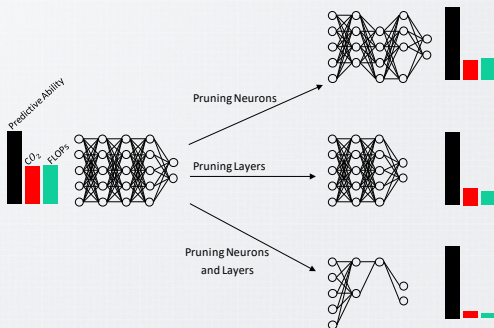
Research Projects

Scalable and Efficient Inference

Compression through Pruning

Compression of Deep Models

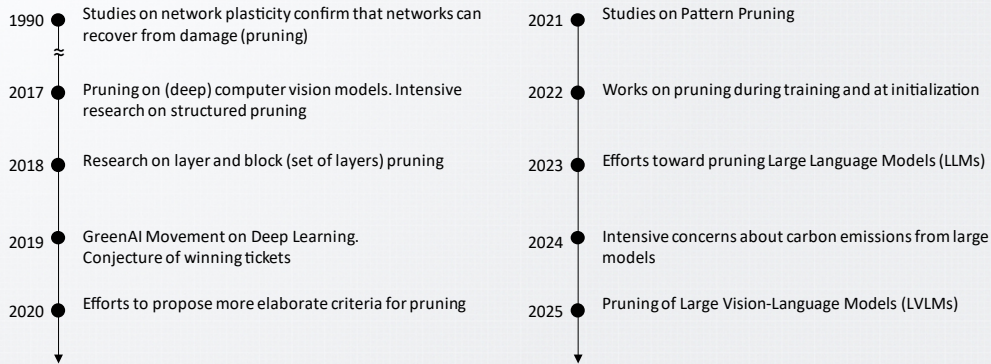
- Among existing solutions to overcome the previous issues, pruning emerges as an effective and hardware-agnostic technique for improving model efficiency
- Pruning involves removing structures (weights, neurons or layers) from the model



Historical Trends

Compression of Deep Models

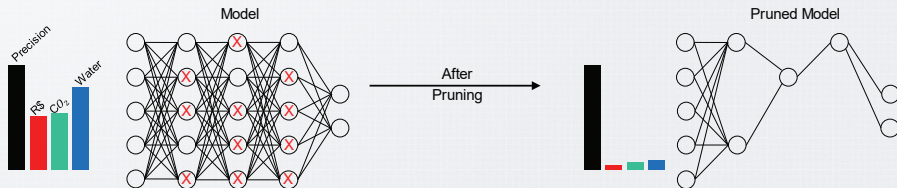
- Noteworthy progress and seminal research in pruning



The Fundamental Question of Pruning

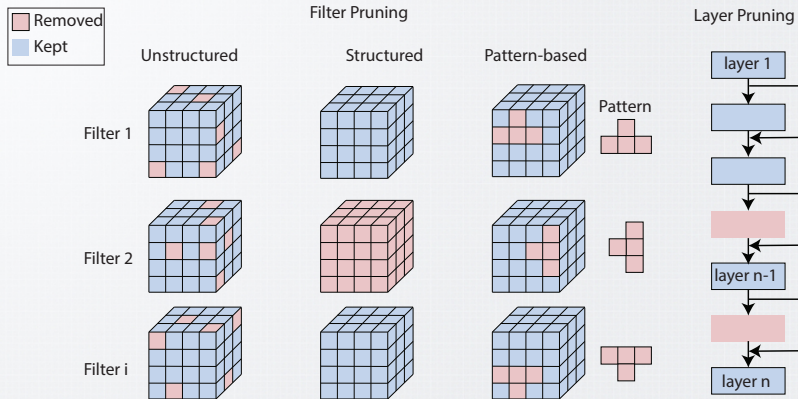
Compression of Deep Models

- An important research question in the field of pruning is *how to decide what structures to remove?*
 - Informally, how to identify the **X** symbol in the figure?



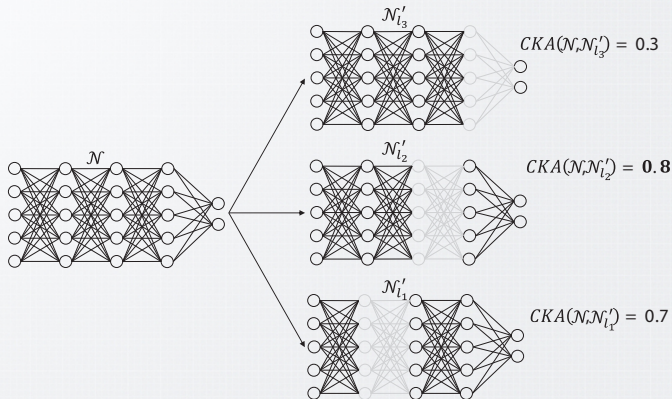
Forms of Pruning

Compression of Deep Models



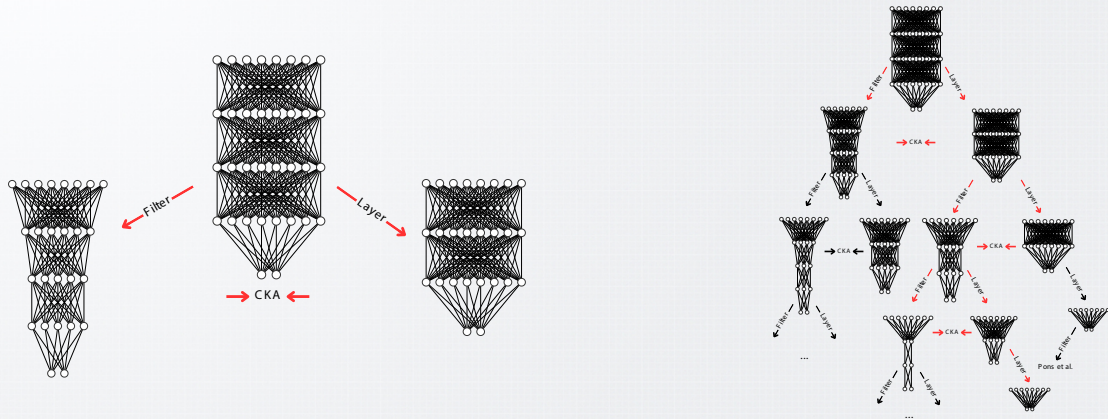
Pruning Layers from Computer Vision Models

Compression of Deep Models



Pruning Neurons and Layers at Once

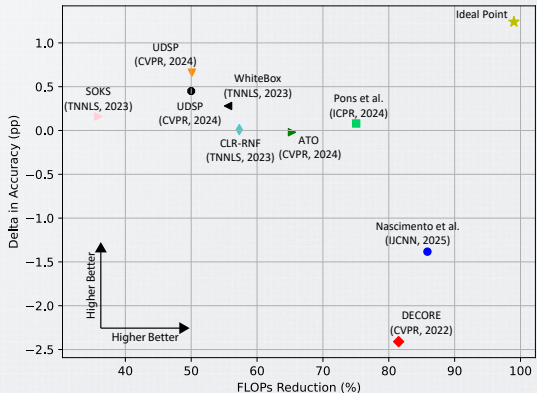
Compression of Deep Models



Pruning Computer Vision Models

Compression of Deep Models

- ResNet56 on CIFAR-10



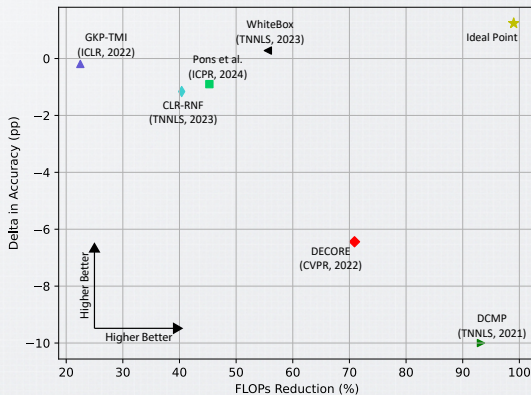
Pons et al. *Effective Layer Pruning Through Similarity Metric Perspective*. ICPR (Oral) and ICML Workshops, 2024

Nascimento et al. *Pruning Everything, Everywhere, All at Once*. IJCNN, 2025

Pruning Computer Vision Models

Compression of Deep Models

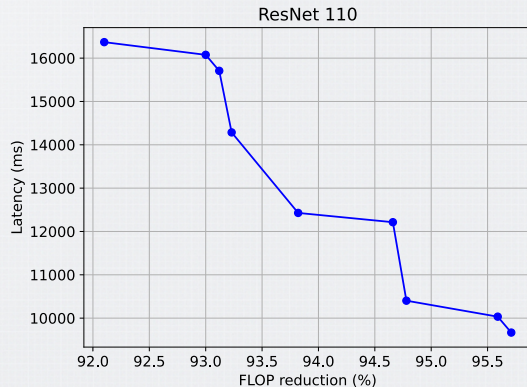
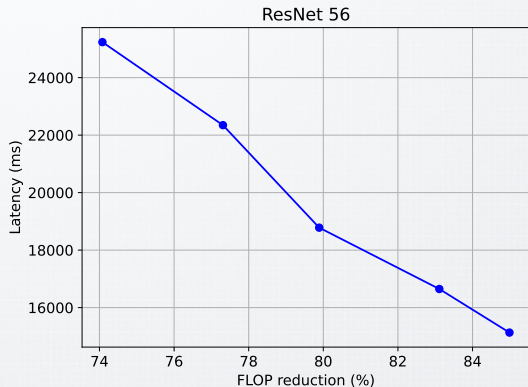
- ResNet50 on ImageNet



Pruning Computer Vision Models

Compression of Deep Models

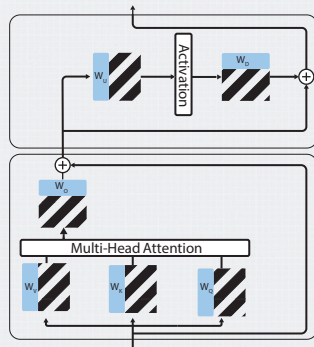
- Deploying pruned models on a resource-constrained ESP32



Pruning Large Language Models

Compression of Deep Models

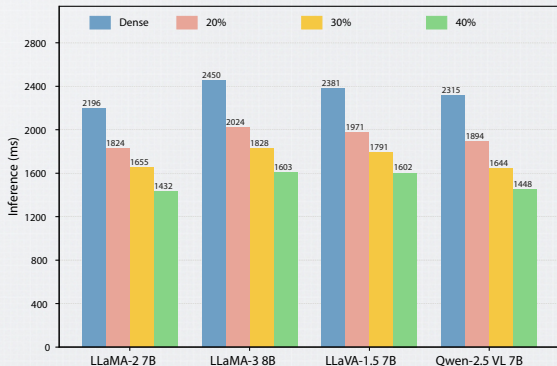
- We propose a simple method that projects input data (X) onto the model weights (W)
 - Thus, it considers the importance of structures in terms of $\|WX\|_1$



Efficient Language, Vision and Multimodal Models

Compression of Deep Models

Compression Ratio	Method	Avg. ↑
0%	LLaMA-2 7B	68.99
30%	SliceGPT (ICLR, 2024)	51.50
	LINEARPATCH (NeurIPS, 2025)	59.39
	CoMe (NeurIPS, 2025)	60.47
	ModHiFi-P (NeurIPS, 2025)	52.44
	PruneNet (ICLR, 2025)	57.08
	AMP (Ours)	61.02
0%	LLaMA-3 8B	72.70
30%	CoMe (NeurIPS, 2025)	52.50
	Yang et al. (ICML, 2025)	54.34
	AMP (Ours)	58.01
0%	LLaMA-3.3 70B	78.25
40%	AMP (Ours)	69.99



Efficient Language, Vision and Multimodal Models

Compression of Deep Models

- Qualitative analysis

Video Demo



Pruning Playground

Compression of Deep Models

- Understanding and evaluating different pruning recipes
 - <https://github.com/arturjordao/TinyTitans>

TinyTitans

Official repository for the tutorial Tiny Titans: Efficient Large Vision, Language and Multimodal Models through Pruning, presented at SIBGRAPI (Tutorial Track)

Basics of Pruning

 [Open in Colab](#)

Pruning Computer Vision Models

 [Open in Colab](#)

Pruning Large Language Models

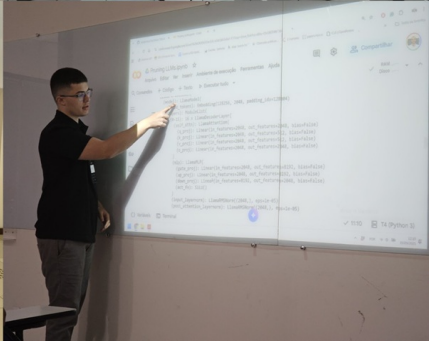
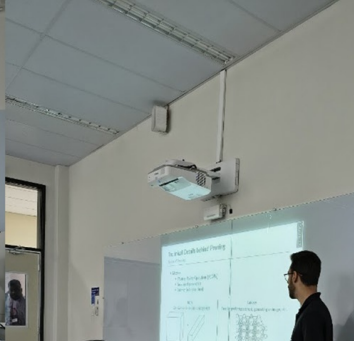
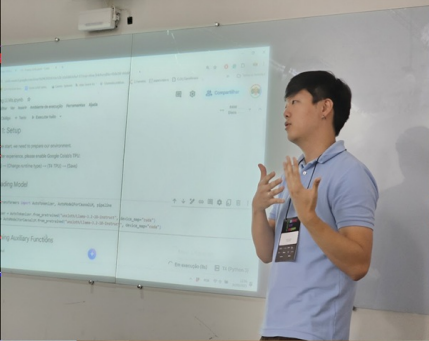
 [Open in Colab](#)

Pruning Detection Model

 [Open in Colab](#)

GitHub





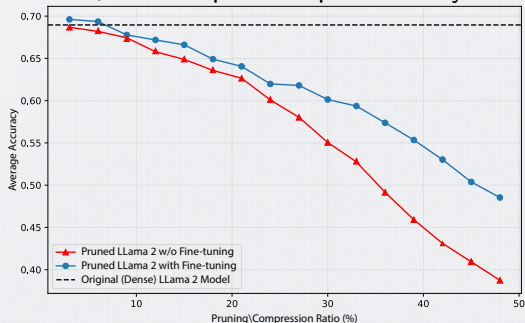
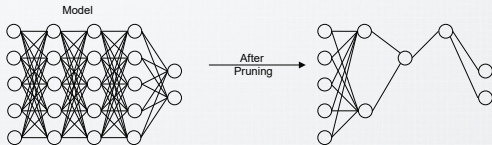
Research Projects

Efficient Training

Motivation

Efficient Training

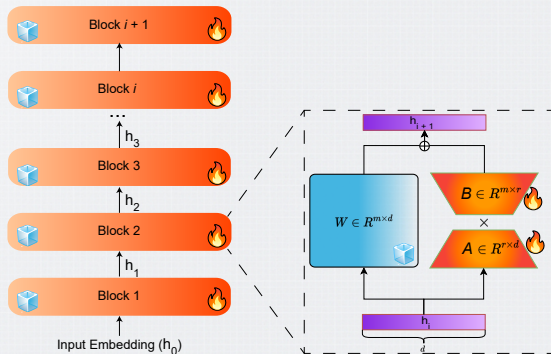
- After the process of pruning, we need to perform some steps of fine-tuning
 - This is important to recover the predictive performance after architecture changes
- If the fine-tuning is computationally prohibitive, the compression process may become a bottleneck



Parameter Efficient Fine-Tuning

Efficient Training

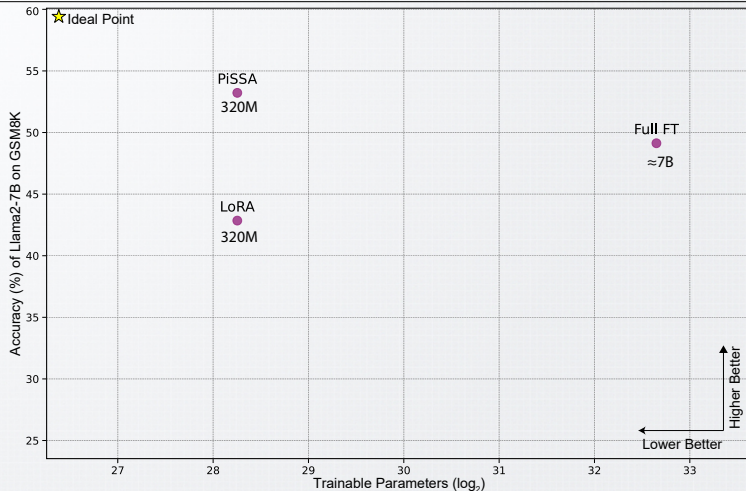
- LoRA¹ constitutes an important technique in the age of large models
 - It allows the exploration of the capacity of these models for specialized tasks with efficient fine-tuning
 - $W + \Delta W \approx W + BA$
- Are all LoRA modules necessary?*



¹Hu et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR, 2022

Parameter Efficient Fine-Tuning

Efficient Training



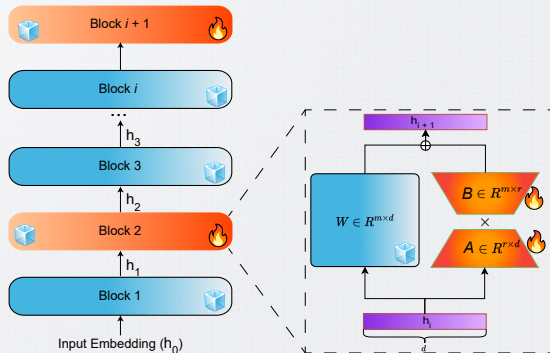
Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR, 2022

Meng et al. *PiSSA: Principal Singular Values and Singular Vectors Adaptation of Large Language Models*. NeurIPS, 2024

Parameter Efficient Fine-Tuning

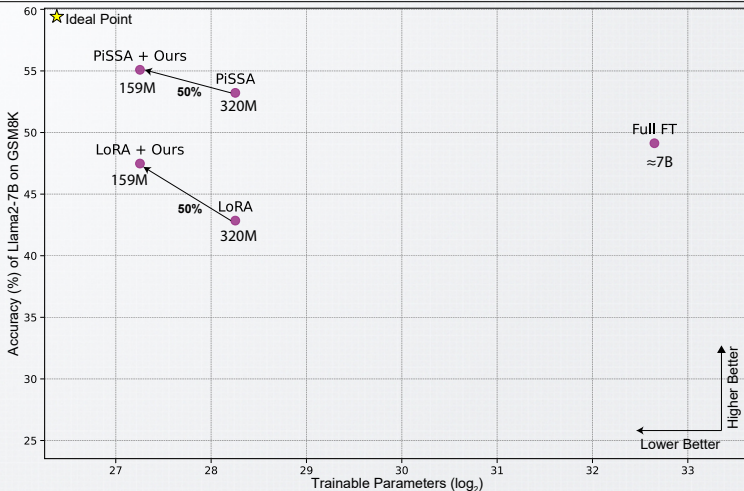
Efficient Training

- *We can effectively choose a subset of transformer layers to fine-tune by measuring their participation in changes on internal representations*
 - This subset comprises a configuration that, when fine-tuned, leads to an effective predictive ability while notably reducing the number of trainable parameters



Parameter Efficient Fine-Tuning

Efficient Training



Meng et al. *PiSSA: Principal Singular Values and Singular Vectors Adaptation of Large Language Models*. NeurIPS, 2024

Ogawa et al. *Layer-wise LoRA fine-tuning: a similarity metric approach*. ICPR, 2026

Parameter Efficient Fine-Tuning

Efficient Training

- Efficient Fine-Tuning
 - Qualitative Results

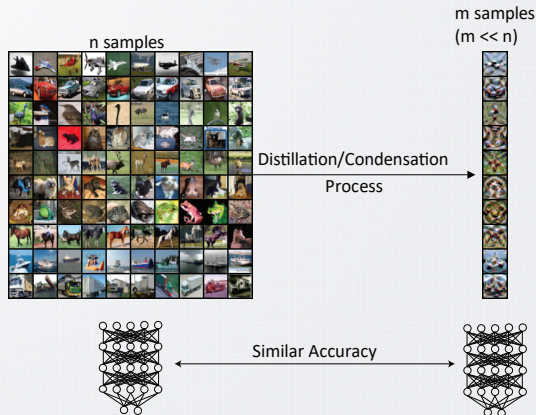
Video Demo



Dataset Distillation

Efficient Training

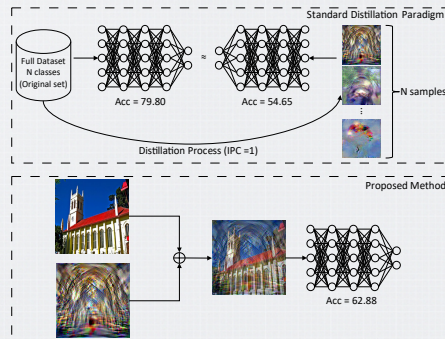
- Dataset distillation focuses on distilling the knowledge of a large dataset into a small, synthetic one



Dataset Distillation

Efficient Training

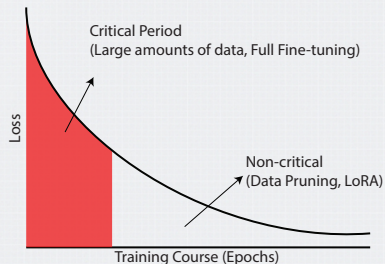
- We propose to mix distilled images with natural images
 - Our method increases predictive performance by up to 11.5 percentage points without increasing the size of the distilled set



Critical Periods

Efficient Training

- Critical Learning Periods comprise an important phenomenon involving deep learning^{1 2}
 - Early epochs play a decisive role in the success of many training recipes
- *How to identify critical periods during the course of training?*
 - In the era of large models, answering this question yields notable gains



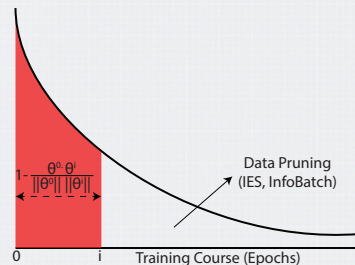
¹Golatkart et al. *Time Matters in Regularizing Deep Networks: Weight Decay and Data Augmentation Affect Early Learning Dynamics, Matter Little Near Convergence*. NeurIPS, 2019

²Achille et al. *Critical Learning Periods in Deep Networks*. ICLR, 2019

Critical Periods

Efficient Training

- *Throughout the course of training, a simple generalization estimation enables successfully identifying when the critical period emerges*
 - Once we establish the end of the critical period, we determine the suitable moment to initiate data pruning
- Our method achieves an average reduction of 33.79% in resource consumption



Efficient Training – Critical Periods

Efficient Training

- Setup
 - Data pruning: Method by Yuan et al.¹ (without critical period)
 - Ours: Data pruning after critical period
 - Baseline: Standard training using full data

Dataset		CIFAR-10			CIFAR-100		
Architecture		ResNet18	ResNet50	VGG16	ResNet34	ResNet101	DenseNet121
IES ¹	Data pruning	94.69	94.64	93.12	77.47	<u>77.99</u>	<u>79.12</u>
	Ours	<u>94.89</u>	<u>94.74</u>	93.31	77.54	77.88	79.30
	Baseline	95.04	94.88	<u>93.29</u>	<u>77.52</u>	78.15	79.07

¹Yuan et al. *Instance-dependent early stopping*. In ICLR, 2025

Fukase et al. *One Period to Rule Them All: Identifying Critical Learning Periods in Deep Networks*. Procedia Computer Science, 2025

Fukase et al. *When to Prune? The Importance of Timing in Data Efficiency Training*. ICPR, 2026

Critical Periods

Efficient Training

- Accuracy delta (ΔAcc) in percentage points and time saved of our method across different datasets using ResNet50

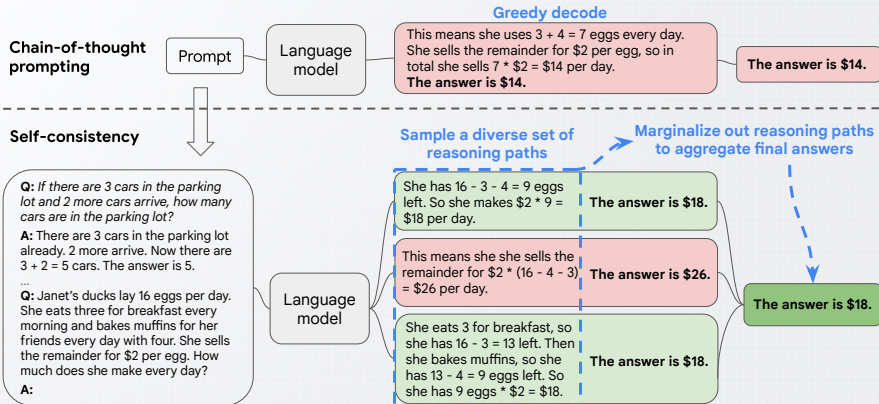
	CIFAR-10	CIFAR-100	EuroSAT	TinyImageNet	ImageNet30
ΔAcc	↓0.32	↓2.36	↑1.02	↓0.20	↑0.34
Saved (%)	49.35	49.47	65.79	60.31	61.70

Research Projects

Test-Time Compute

Self-consistency

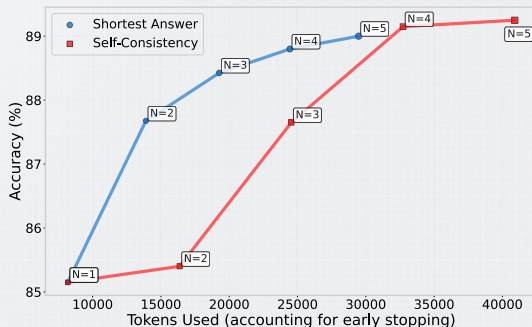
Test-Time Compute



Self-consistency

Test-Time Compute

- Recent literature shows that reasoning models suffer from overthinking¹
 - A phenomenon in which they generate more tokens than necessary
- We demonstrate that the simple and counter-intuitive heuristic of selecting the shortest solution is highly effective²



¹Chen et al. *Do NOT think that much for $2+3=?$ on the overthinking of long reasoning models*. ICML, 2025

²Dinardi et al. *The Virtues of Brevity: Avoid Overthinking in Parallel Test-Time Reasoning*. NeurIPS Workshop on Efficient Reasoning, 2025

Self-consistency

Test-Time Compute

Model	Method	AIME	LiveCodeBench
DeepSeek-R1	Individual Attempts	85.0%	76.5%
	Shortest Solution	89.0%	79.2%
	Self-Consistency	89.2%	*
	Longest Solution	78.2%	76.5%
Grok-3-mini	Individual Attempts	81.0%	69.5%
	Shortest Solution	85.2%	69.2%
	Self-Consistency	86.2%	*
	Longest Solution	74.9%	66.8%
Qwen3-32B	Individual Attempts	89.5%	78.6%
	Shortest Solution	92.5%	79.5%
	Self-Consistency	93.0%	*
	Longest Solution	85.5%	76.8%

Procedural Benchmark for Multimodal Graph Reasoning

Test-Time Compute

- Several multimodal suites barely rely on their images
 - We re-evaluate an open MLLM on twelve popular benchmarks after shuffling the pixels of every input image: it destroys the picture and leaves the question untouched

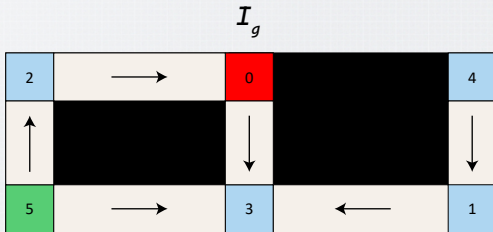
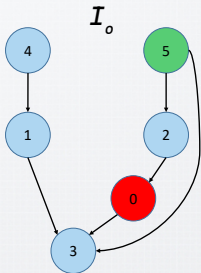
Benchmark	Original	Shuffled	Δ
DocVQA	87.8	7.1	80.7
OCRBench	75.7	1.2	74.5
TextVQA	79.5	7.0	72.5
ChartQA	72.3	9.9	62.4
InfoVQA	60.3	17.8	42.5
OK-VQA	44.2	4.5	39.7
GQA	60.0	34.9	25.1
RealWorldQA	59.2	38.6	20.6
MMStar	42.6	26.9	15.7
AI2D	70.0	55.4	14.6
ScienceQA	76.7	64.7	12.0
CMMMU	35.9	32.7	3.2

Procedural Benchmark for Multimodal Graph Reasoning

Test-Time Compute

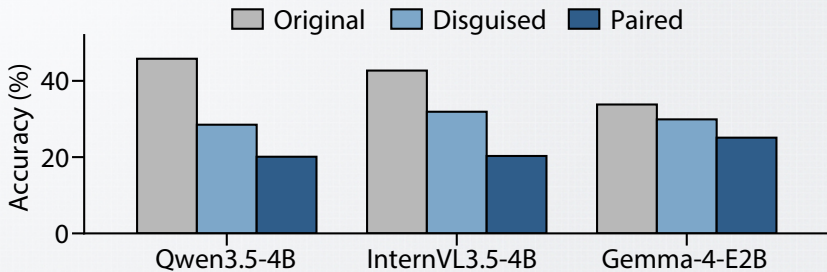
Require: *seed, task, difficulty*

- 1: $G \leftarrow \text{SAMPLEGRAPH}(\text{seed}, \text{task}, \text{difficulty})$
- 2: $\text{answer} \leftarrow \text{SOLVE}(\text{task}, G)$
- 3: $(I_o, P_o) \leftarrow \text{RENDERORIGINAL}(G, \text{task})$
- 4: $(I_g, P_g) \leftarrow \text{RENDERDISGUISE}(G, \text{task}, \text{seed})$
- 5: **return** $(I_o, P_o), (I_g, P_g), \text{answer}$



Procedural Benchmark for Multimodal Graph Reasoning

Test-Time Compute



Resources

- Machine Learning Lectures
 - <https://github.com/arturjordao/Machine-Learning/>
- Deep Learning Lectures
 - <https://github.com/arturjordao/PCS5022-DeepLearning>
 - Videos available
- Scientific Writing (**my writing style**)
 - <https://github.com/arturjordao/PaperWriting>
 - Videos available
- LaTeX-based presentation slides
 - <https://www.youtube.com/watch?v=5fptqcrSBg4>

1. Fukase et al. *When to Prune? The Importance of Timing in Data Efficiency Training*. ICPR, 2026
2. Ogawa et al. *Layer-wise LoRA fine-tuning: a similarity metric approach*. ICPR 2026

